

Arpan Pathak

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SUMMARY

Software Engineer with 7+ years architecting high-scale distributed systems, ML-driven platforms, and scalable microfrontend UIs at Amazon, Microsoft, and Oracle. Proficient in Go, Rust, Kubernetes, and cloud-native infrastructure. Built ad ranking engines driving \$5.4M monthly revenue and led sovereign cloud expansions with eBPF/Cilium.

EXPERIENCE

Software Developer II | Microsoft | Redmond, Washington | June 2025 - Present

- Designed and deployed eBPF-based network policies using Cilium for **Microsoft Purview's sovereign** regions. Engineered the control plane to enforce L7 filtering based on **geographic traffic origins**, and authored custom **Kubernetes Operators** to reconcile compliance states automatically across **AKS clusters** in France and Germany.
- Built an AI-driven policy engine that auto-generates Cilium network policies and scans for misconfigurations, **reducing manual policy review cycles from 4 hours to near real-time**, ensuring zero policy drift across France/Germany sovereign regions.
- Spearheaded the **C++-to-Rust** migration for the Information Protection service, shifting memory safety to compile-time. **Eliminated use-after-free and buffer-overflow CVEs** in the audited path, a critical win for DORA compliance, while maintaining performance parity.

Senior Software Engineer | Oracle Cloud Infrastructure | Seattle, Washington | October 2024 - March 2025

- Engineered Terraform provider for Oracle Autonomous Database, replacing legacy control plane infrastructure with potential \$1.2M/month cost savings.
- Integrated OCI Security Vault across control plane services, establishing secure secret management and automated key rotation protocols.
- Architected low-latency internals for a proprietary database engine, optimizing lock-free data structures to reduce contention in high-concurrency read-write paths, improving transaction throughput by 45,000 ops/s.

Software Development Engineer II | Amazon | Seattle, WA & Hyderabad, India | October 2021 - October 2024

- Scaled Advertisement Sponsorship Campaign Management Tool across **10 Geographic Regions** and integrated additional features such as real-time Event Bus & Audit Trail for **Programmatic Guarantee Advertising**, achieving **100ms latency** while processing millions of ad inventory deals per second.
- Reduced over-delivery of deals impression by roughly **30%** and budget overspending by **25%** through real-time monitoring and automated controls.
- Architected ensemble model pipeline combining **XGBoost** and **Deep Learning** for real-time ads & deal ranking, driving **5% CTR increase**, **30%** higher deal impressions, and **\$5.4M monthly revenue** growth across multi tenant publisher channels such as **FreeVee, Twitch, MiniTV, Prime Videos, Amazon Retail**.
- Designed and delivered a uniqueness constraint indexing solution and SDK for a multi-tenant NoSQL data store using two-phase commit to support list-type fields within a publisher tenant, achieving serializability (the highest level of isolation) while handling 3k writes per second.
- Developed **BERT-based conversational AI** model and UI component that reduced Amazon WorkEvent support ticket triage **SLA from 7 days to 2 hours** while handling **5,000 QPS** at peak capacity.
- Led migration of WorkEvents UI to React.js microfrontend architecture with AWS CloudFront; improved developer experience, feature shipping time, and approval processes to enable full team autonomy while reducing deployment cycles from **5 days to 30 minutes**.
- Engineered and fine-tuned Spark big data jobs to process millions of daily WorkEvent schedules and punch data, consolidating streams into Amazon Redshift for OLAP use cases with automated BIOps dashboards using AWS QuickSight.
- Implemented a real-time in-memory cache and incremental data refresh mechanism for AWS QuickSight BI Datasets to accelerate sorting and filtering queries for higher data freshness and reducing data staleness from **1 hour to 5 minutes**.

Software Development Engineer | Razorpay | Bengaluru, India | August 2020 - February 2021

- Built UPI Payment Gateway Aggregator using Golang and Protobuf, processing 1M+ hourly transactions and generating Rs.1.2 Cr revenue from payment fees.
- Engineered refund and reconciliation systems, enhancing transaction transparency and eliminating manual reconciliation overhead.
- Optimized payment infrastructure maintaining 99.99% uptime with 150ms P99 latency under 50,000 TPM peak loads.

Software Engineer | Mindfire Solutions | Bhubaneswar, India | August 2018 - February 2020

- Developed full-stack workflow tools and AR-based product search microservice, serving 10,000+ daily queries with 2x performance improvement.
 - Built administrative dashboards that reduced data retrieval latency from 5 minutes to 1 minute for 50+ team members.
 - Implemented WebSocket-based multipart file upload system handling 1,000+ daily uploads for 1,000+ concurrent users.
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SKILLS

Languages: Rust, Kotlin, Java, Go, Python, TypeScript, C++, C#, Shell Script

Distributed Systems & Cloud: AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS, Bicep), Kubernetes (EKS, AKS), Docker, Terraform, AWS CDK, Apache Kafka, Apache Spark, Apache Flink, Kubebuilder, Istio, Cilium, Kubernetes Gateway API

Distributed Data & Storage: DynamoDB, PostgreSQL, Redis, MongoDB, Amazon Redshift, Amazon S3

Frontend Engineering: React.JS, Javascript, Angular, HTML 5, SCSS

Observability: AWS CloudWatch, Prometheus, Grafana, Azure Monitor

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, XGBoost, Transformer Models (BERT), Recommendation Systems, Deep Learning, Neural Networks, MLOps, Natural Language Processing(a.k.a Computer Linguistics), Computer Vision, Audio Signal Processing

Core CS Fundamentals: Data Structures & Algorithms, Distributed Systems (consensus, replication, sharding), Concurrency, System Design, Networking (TCP/IP, UDP, TLS, DNS, HTTP 3, load balancing)

EDUCATION

Bachelor of Technology in Computer Science & Engineering | RCC Institute of Information Technology

Kolkata, India | 2018
